

Ruihe Zhang

📍 Montreal, QC 📞 (514) 654-4081 ✉️ ruihe.zhang@mail.mcgill.ca [LinkedIn](#) [GitHub](#) [Website](#)

EDUCATION

McGill University, Desautels Faculty of Management

Master of Management in Analytics | GPA: 3.74/4.00

Montreal, QC

Jun 2025 – Feb 2027 (Expected)

- Graduate Entrance Scholarship; Scope-AI-Thon 2025 Global 2nd Prize; Microsoft Certified: Azure Fundamentals (AZ-900); Databricks Fundamentals Accreditation.

Hefei University of Technology

Bachelor of Economics in International Economics and Trade | GPA: 3.68/4.00 | GRE: 328

Hefei, China

Sep 2021 – Jun 2025

- English-medium exchange at Hanyang University, South Korea (GPA: 4.15/4.50); awards in national innovation and provincial trade competitions.

PROFESSIONAL EXPERIENCE

Scotiabank – McGill Analytics Capstone

Lead Modeller, Credit Risk Forecasting & Reporting

Montreal, QC

Sep 2025 – Apr 2026

- **Risk Data Reconciliation:** Led a Provision for Credit Losses workstream covering Canada's five largest banks; reconciled quarterly figures from original **PDF and CSV disclosures** into a traceable 2017Q4–2025Q4 dataset.
- **Forecast Validation:** Built and rolling-backtested time-series and machine-learning models in **R and Python**; added macroeconomic indicators that reduced validation MAPE by **55%** versus a statistical baseline.
- **Risk Reporting:** Documented assumptions, exceptions, and scenario results and delivered them through **Power BI and Streamlit**, supporting evidence-based review by client and faculty stakeholders.

Department of Earth & Planetary Sciences, McGill University

Research Assistant, Data Quality & Predictive Analytics

Montreal, QC

May 2026 – Aug 2026

- **Multi-Source Data Engineering:** Integrated **130,176 five-minute GNSS–IR observations** with 434 daily arc files, pressure records, CHS tides, and ECCO weather; enforced data contracts for timestamps, coverage, duplicates, plausibility, and provenance.
- **Model Benchmarking & Tuning:** Compared **statistical modelling, machine-learning, and neural-network approaches**—regularized regression, tree ensembles, GPR, EDM, and MLP—using **expanding-window validation** and seeded **Optuna** tuning.
- **Explainable Model Selection:** Used confusion matrices, class-specific metrics, **permutation importance**, and **SHAP** to explain performance; selected a parsimonious four-feature **Capped Exact GPR** with **0.162 m later-holdout RMSE**, reducing error by **6.11 cm** versus a UTC-aligned tide model with a vertical-reference offset.

Agricultural Bank of China

Account Manager Assistant

Nanjing, China

Jan 2023 – Feb 2023

- **Credit Due Diligence:** Supported due diligence on **10+ commercial loan files** by reviewing client financial statements and industry information to assess repayment capacity and credit risk.
- **Documentation Control:** Organized supporting records, documented discrepancies, and summarized findings for account-manager review while handling confidential client information.

PROJECTS

BIXI Demand Forecasting Application | AWS, FastAPI, Docker, MLflow

Jan 2026 – Mar 2026

- **Cloud Infrastructure & Orchestration:** Co-designed an eight-stage cloud ML workflow using **AWS CDK, Batch, S3, and MLflow**; incorporated validation, explanation, fairness, drift monitoring, and versioned model registration.
- **Secure API Deployment & CI/CD:** Developed **FastAPI** endpoints and Streamlit interfaces; implemented **IAM**-based artifact access, pytest checks, and resumable S3 markers for controlled inference and recovery.

Community Resource-Matching Application | Scope-AI-Thon 2025 Global 2nd Prize

Mar 2025

- **Privacy-Conscious AI Product:** Built a Python application that matched shelter services to user needs and generated multilingual, low-literacy guides from structured resource data.

Pokémon Analytics | Web App | Power BI Dashboard

Jan 2026

- **Data Modelling & BI:** Normalized nested REST API data into linked tables and built a **Power BI star schema** with reusable **DAX measures**, disconnected comparison tables, and side-by-side What-If analysis.

SKILLS & PORTFOLIO

Banking & Analytics: Credit risk, due diligence, data reconciliation, statistical modelling, machine learning, deep learning, model validation, project coordination, risk reporting, Excel, Power BI (DAX, Power Query)

Technical: Python (Pandas, scikit-learn, PyTorch, FastAPI), R, SQL, AWS, Azure, Databricks, Docker, MLflow, Git/GitHub Actions, Linux, LaTeX

Power BI: Global Sales | PokéStats | Credit Risk Provision | Earnings-Call Sentiment

Applications: PCL Forecast | GNSS | BIXI Demand | Scholar | Typhoon Vision | DISSA | Pokémon Web | Gesture CV | Minesweeper

Languages & Interests: English (Fluent), Mandarin (Native), French (Basic); Chinese Go (Amateur 1st Dan), badminton